Vaibhav Verma

Generative AI Engineer

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Professional Summary

Python and Generative AI Engineer focused on building LLM-powered applications using FastAPI, OpenAI APIs, and LangChain. Skilled in RAG pipelines, embeddings, vector search, tool calling, and LLM application workflows. Experienced in developing AI-integrated applications, debugging code, and deploying web services.

Technical Skills

Programming: Python
Backend: FastAPI, REST APIs, API Integration, Streaming Responses
Generative AI: OpenAI API, LangChain, Prompt Engineering, Runnables & Chains, Tool Calling, AI Agents
RAG & Retrieval: Document Loaders, Text Splitters, Embeddings, Pinecone Vector Database, Semantic Search, Reranking, RAG Evaluation
Frontend & Tools: React, AI-assisted UI Development, AI-assisted Code Review & Debugging, Git, GitHub, VS Code, Vercel, Render

Projects

ORYN - Enterprise Knowledge RAG Engine

Tech Stack: React.js, TypeScript, Python, FastAPI, PostgreSQL, Pinecone, OpenAI API, Cohere
GitHub | Live

- Developed an AI-powered document platform using RAG to enable intelligent question-answering from uploaded documents.
- Implemented hybrid search using Pinecone vector search and PostgreSQL full-text search for relevant document retrieval.
- Integrated OpenAI and Cohere for AI-generated responses, embeddings and result reranking.
- Built secure multi-tenant workspaces with authentication, role-based access, Google OAuth and audit logging.
- Implemented real-time response streaming, document processing, source citations and evaluation analytics.

NEXIS - AI Agent & Workflow Automation Platform

Tech Stack: React.js, JavaScript, Python, FastAPI, PostgreSQL, LangGraph, OpenAI API
GitHub | Live

- Developed an AI-powered platform to create autonomous agents and automate multi-step tasks using LangGraph.
- Integrated OpenAI tool-calling to enable AI agents to interact with external tools and execute tasks.
- Implemented custom REST tools, API key management and multi-step workflow automation.
- Built real-time execution monitoring using Server-Sent Events (SSE), with execution history and event tracking.
- Implemented secure authentication, role-based resource ownership, approval-based tool execution and API security.

SYNAXYS – AI-Powered Code Review & GitHub Intelligence Platform

Tech Stack: React.js, JavaScript, Python, FastAPI, PostgreSQL, LangChain, OpenAI API, Pinecone, GitHub API
GitHub | Live

- Developed an AI-powered code review platform to analyze GitHub pull requests and identify potential bugs, code quality issues, and security concerns.
- Integrated GitHub OAuth, GitHub App, and webhooks to connect repositories and automatically trigger code reviews.
- Implemented LangChain and OpenAI-powered review workflows to generate structured, evidence-backed findings with file references, line numbers, severity, and confidence levels.
- Integrated Pinecone vector database for semantic code search and retrieval of relevant code context across repositories.

- Built an interactive dashboard with code diff visualization, custom review rules, finding management, and human-approved GitHub comment publishing.

Work Experience

Frontend Developer Intern | ABYM Technologies

Noida, Sector 62 | Jun 2025 – Nov 2025

- Completed hands-on training in frontend development using HTML5, CSS3, JavaScript, and React.js.
- Practiced building responsive user interfaces using modern CSS techniques and reusable React components.
- Gained practical knowledge of React fundamentals, including components, props, state, and React Hooks.
- Used Git and GitHub to understand version control and practice managing code changes during development.

Education

Bachelor of Technology (B.Tech) in Computer Science & Engineering

Chaudhary Devi Lal State Institute of Engineering & Technology, Sirsa, Haryana

2022 – 2026 | CGPA: 7.5/10